

# NEXUS MIND VAULT â FUNCTIONAL WALKTHROUGH CATALOG

Master Scenario-Driven Test Specifications (End-to-End Verification Manual)

## [ SUITE 01 ] AUTHENTICATION & IDENTITY LIFECYCLE

### TC-01.1: Federated Google Sign-In & Session Persistence

*Prerequisites: User logged out, server active on http://localhost:3000*

#### Execution Steps:

1. Navigate to http://localhost:3000.
2. Verify Login Page renders with title "Nexus Mind Vault" and "Continue with Google" button.
3. Click [ Continue with Google ] (or enter local sandbox session).

#### Expected Assertions:

- â User profile avatar and email render in the top navigation header.
- â Application smoothly transitions to the Journal workspace view.
- â localStorage item "vault\_journal\_session\_user" contains valid user UID and token.

### TC-01.2: Session Sign Out & State Purge

*Prerequisites: User authenticated in active session*

#### Execution Steps:

1. Click the user avatar in the top-right header.
2. Click [ Sign Out ] button.

#### Expected Assertions:

- â User session is completely removed from local storage and memory.
- â Vault mode immediately resets to "protected" cover mode.
- â UI redirects to the Login Page.

## [ SUITE 02 ] SECRET GATE & SOVEREIGN DECRYPTION SEQUENCE

### TC-02.1: Secret Gate Trigger via Logo Long-Press (750ms)

*Prerequisites: User signed in, app in Protected Cover Mode*

#### Execution Steps:

1. Position cursor on the top-left NexusMindVaultLogo.
2. Press and hold mouse button / touch for >= 750 milliseconds.

#### Expected Assertions:

- â Haptic / visual glow trigger emits on the logo perimeter.
- â The "Unlock Real Vault" modal appears with input for Secret Passphrase.

### TC-02.2: Master Secret Code Verification & Decryption Sequence

*Prerequisites: Unlock modal open*

#### Execution Steps:

1. Enter configured Master Secret Code (e.g. 123456).
2. Click [ Unlock Real Vault ] or press Enter.

#### Expected Assertions:

- â Toast displays "Secret Code Verified: Nexus MIND Vault Unlocked."
- â Cinematic Decryption Sequence Modal runs diagnostic cryptographic checks.
- â Top HUD updates to "Sovereign Enclave Active" with solid high-contrast badges.
- â Session verification hash (e.g. 0xEF835A0E...) renders in the top HUD.

## [ SUITE 03 ] DURESS ANTI-COERCION PROTOCOL

### TC-03.1: Coerced Physical Challenge Unlock via Duress PIN

*Prerequisites: Duress PIN configured in Settings (e.g., 999888)*

#### Execution Steps:

1. Trigger Secret Gate logo long-press to open Unlock Modal.
2. Enter Duress PIN (999888) instead of Master Secret Code.
3. Click [ Unlock ].

#### Expected Assertions:

- â Zero Error / Zero Alarm: Modal closes smoothly without giving away duress state.
- â Mounts the harmless Decoy Parallel Persona (Botanical Field Journal).
- â Real journals, crypto capsules, and emergency directives remain 100% sealed.

## [ SUITE 04 ] MULTIMODAL JOURNAL STUDIO

### TC-04.1: Multi-Mood Reflection with Folder Organization

*Prerequisites: Real Vault unlocked, Journal view active*

#### Execution Steps:

1. Select Write tab in Journal view.
2. Enter Title: "Autonomous Security Architecture".
3. Enter Content: "Testing zero-trust cryptographic isolation and policy enforcement."
4. Select Mood: "Focused" (green pill), Folder: "Architecture".
5. Click [ Save Entry to Vault ].

#### Expected Assertions:

- â Toast displays "Reflection encrypted & saved to Real Vault."
- â New reflection appears at top of timeline with #architecture tag and Focused pill.

### TC-04.2: Multimodal Camera Studio Snapshot & Lens Filter

*Prerequisites: Journal composer active*

#### Execution Steps:

1. Click Camera icon to open Innovative Camera Studio.
2. Select Lens Filter (e.g. Cyberpunk Emerald).
3. Click [ Capture Snapshot ].

#### Expected Assertions:

- â Snapshot is converted to isolated Base64 data attachment.
- â Image preview thumbnail displays under composer draft.

### TC-04.3: Talk with AI Cognitive Mirror (Multi-Turn Chat)

*Prerequisites: Journal view active*

#### Execution Steps:

1. Click Talk (Chat with AI) tab.
2. Type prompt: "Analyze my dominant cognitive theme across recent reflections." and click Send.

#### Expected Assertions:

- â User message renders in chat with timestamp.
- â Gemini AI model generates cognitive synthesis response.
- â Subsequent query turns maintain full multi-turn conversational history.

## [ SUITE 05 ] FLOATING AMBIENT AUTO-LOCK WARNING BANNER

### TC-05.1: 30-Second Warning Alert & Progress Bar Depletion

*Prerequisites: Real Vault active, inactivity timer reaches <= 30 seconds*

#### Execution Steps:

1. Allow timer to countdown to 0:30 without typing or clicking.

#### Expected Assertions:

- â Floating Ambient Action Banner slides into top-center viewport with red glow.
- â Audio warning pulse chime sounds.
- â Live progress micro-bar visually depletes second-by-second in real-time.

### TC-05.2: Keyboard Shortcut Ctrl+E Session Extension

*Prerequisites: 30-second warning banner visible on screen*

#### Execution Steps:

1. Press Ctrl + E on the keyboard (or click [ +5m / Extend ] button).

#### Expected Assertions:

- â Warning banner dismisses immediately.
- â Toast displays "Session extended."
- â HUD auto-lock countdown timer resets to full selected duration.

## [ SUITE 06 ] SOVEREIGN SECURITY HUD & SESSION SYNC

### TC-06.1: Dynamic Timeout Synchronization Across Settings

*Prerequisites: Real Vault unlocked*

#### Execution Steps:

1. Navigate to Settings -> Auto-Lock & Inactivity.
2. Select 15 min or 30 min button.

#### Expected Assertions:

- â HUD timer in header immediately synchronizes to 15:00 or 30:00.
- â Extend button dynamically updates to +15m or +30m.

## TC-06.2: One-Click Panic Purge (Instant Lock)

*Prerequisites: Real Vault unlocked*

### Execution Steps:

1. Press ESC key twice or click red [ Panic Lock ] button in HUD.

### Expected Assertions:

- â Panic sound plays immediately.
- â Real Vault locks instantly, purging decrypted state from memory.
- â UI returns to Protected Cover Mode.

## [ SUITE 07 ] SEMANTIC MEMORY GRAPH & NEURAL VAULT CHAT

### TC-07.1: D3 Force-Directed Graph Interaction & Node Clusters

*Prerequisites: Real Vault unlocked*

### Execution Steps:

1. Click Memory Graph (Nexus Mind) tab in navigation header.
2. Click on an entity node (e.g. "Architecture").

### Expected Assertions:

- â Graph dynamically highlights connected relationship links and vertices.
- â Semantic inspection card displays connected entries count and summary.

### TC-07.2: Subtab Switch to Neural Vault Chat (Material 3 Styling)

*Prerequisites: Nexus Mind view active*

### Execution Steps:

1. Click [ Neural Vault Chat ] segmented pill button.

### Expected Assertions:

- â Button active state highlights in solid Google Blue (#1a73e8) with white text.
- â Neural chat conversation pane renders with multi-turn history.

## [ SUITE 08 ] SEALED TIME CAPSULES & CRYPTOGRAPHIC TIMELOCKS

### TC-08.1: Create Sealed Time Capsule with SHA-256 Checksum Seal

*Prerequisites: Real Vault unlocked*

### Execution Steps:

1. Navigate to Time Capsules -> click [ + Create Sealed Time Capsule ].
2. Enter Title: "Quantum Resilience Directive 2028".
3. Set unlock date to 14 days in future; select Mood: "Calm".
4. Click [ Cryptographically Seal Capsule ].

### Expected Assertions:

- â Capsule is sealed with immutable SHA-256 integrity hash.
- â Capsule renders as "Sealed" with live countdown; message content masked until unlocked.

## [ SUITE 09 ] NEXUS LEGACY GUARDIAN MULTI-POLICY PROTOCOL

### TC-09.1: Deploy Multi-Category Emergency Directives

*Prerequisites: Real Vault unlocked*

### Execution Steps:

1. Navigate to Future Capsules -> click [ + Add Legacy Policy ].
2. Category: Family Medical Living Will; Check-In: 72h; Grace: 24h.
3. Add trusted contact email -> click [ Deploy Policy ].

### Expected Assertions:

- â Policy is saved and sealed with status "Active".
- â Check-in deadline countdown timer starts running.

### TC-09.2: Global Proof-of-Life Pulse (Reset All Policies)

*Prerequisites: Legacy policies active*

### Execution Steps:

1. Click [ Global Proof-of-Life Pulse (Reset All Policies) ].

### Expected Assertions:

- â Audio pulse chime plays.
- â All active policies refresh their check-in deadlines to full duration.
- â Toast displays "Global Heartbeat confirmed across X policies!"

## [ SUITE 10 ] AUTOMATIC PROOF-OF-LIFE ON SECRET CODE UNLOCK

### TC-10.1: Automatic Heartbeat Trigger on Secret Code Unlock

*Prerequisites: Legacy policy active with elapsed time*

#### Execution Steps:

1. Lock the vault -> enter Secret Code in Secret Gate to unlock.

#### Expected Assertions:

- â Unlock sequence automatically invokes recordGlobalHeartbeatPulse().
- â Proof-of-Life heartbeat is registered without needing manual clicks.

## [ SUITE 11 ] ZERO-TRUST CRYPTOGRAPHIC ASSURANCE AUDIT

### TC-11.1: Run Live In-Browser Cryptographic Self-Audit Benchmark

*Prerequisites: Real Vault unlocked*

#### Execution Steps:

1. Navigate to Settings -> scroll to Zero-Trust Cryptographic Assurance.
2. Click [ Run Live Cryptographic Audit ] button.

#### Expected Assertions:

- â Auditing animation displays for ~1s while testing CSPRNG entropy and storage.
- â Diagnostic panel renders "Live Audit Result: 100% Cryptographic Health Confirmed".
- â Benchmark passes: Entropy Valid (256-bit), Zero Plaintext Leak, Duress Isolated.

## [ SUITE 12 ] MASTER SECURITY CREDENTIALS CONFIGURATION

### TC-12.1: Update Master PIN & Secret Code

*Prerequisites: Settings view active*

#### Execution Steps:

1. In Master Security Credentials card, click [ Change Credentials ].
2. Enter Current PIN, New 6-digit PIN, Confirm PIN, and New Secret Passphrase.
3. Click [ Update & Re-Encrypt ].

#### Expected Assertions:

- â Toast displays "Master Vault credentials successfully updated and re-encrypted."
- â Unlocking with old PIN fails; unlocking with new PIN succeeds.

## [ SUITE 13 ] GLOBAL COMMAND PALETTE (CTRL+K / CMD+K)

### TC-13.1: Global Keyboard Navigation & Tab Routing

*Prerequisites: Any active view in application*

#### Execution Steps:

1. Press Ctrl + K (or Cmd + K on macOS).
2. Type "Time Capsules" and press Enter.

#### Expected Assertions:

- â Command palette modal opens immediately with dark backdrop.
- â Selecting option smoothly routes directly to Time Capsules view.

## [ SUITE 14 ] HIGH-CONTRAST DARK & LIGHT THEME SWITCHING

### TC-14.1: Theme Mode Switching & Solid Text Contrast

*Prerequisites: Header visible*

#### Execution Steps:

1. Click Theme Toggle icon (Moon / Sun) in top header.

#### Expected Assertions:

- â Application toggles between Security Dark (#0b0f19) and System Light (#f8fafc).
- â All tags and badges render with solid high-contrast text across both themes.